

Assessment of Thermodynamic variables affecting phase nucleation

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ABSTRACT

Complete understanding of any natural phenomenon can only be achieved by explicitly expressing it in terms of primitive variables. The laws of thermodynamics have been extensively studied for centuries, accelerating societal progress through developments across several branches of science and technology. Phase nucleation can be observed in the melting and freezing of rocks and ice caps, abundant shapes of snow flocks, rain regimes, snowfall, casting, and metal forming, among other applications. The most challenging aspect of nucleation is the equilibrium shift controlled by the gradients in temperatures, species, work, and other variables, in accordance with the first law of thermodynamics, which defines the thermal field. In this study, the thermal field and thermal field tensor were defined in terms of equilibrium dislocations to assess their primitive variables. Theoretical simulations for the Gibbs–Thomson coefficients were applied to predict the experimental microstructure evolution.

1. Introduction

Nucleation can be defined as a general phenomenon applied in several natural and industrial processes, such as crystallization, melting, boiling, condensation, and glass transition [1]. Recently, nucleation has been found to be applicable in rapid prototyping additive manufacturing [2]. It is used in analyses ranging from classical and statistical thermodynamics to molecular dynamics and quantum theory [3]. Classical nucleation theory (CNT) originated from Fahrenheit's supercooling of water [4] and later received thermodynamic support [5,6]. CNT is useful for understanding the evolution of phase nucleation through thermodynamic variables and provides valuable theoretical information for numerous practical applications. According to Nanev [7], CNT provides the basis for understanding crystal nucleation and is not obsolete. Its use is inevitable for elucidating crystal nucleation via melt crystallization. However, it fails to predict vapor–liquid systems [8]. Density functional theory (DFT) is a nonclassical nucleation theory based on a density function approach concerning the intrinsic free energy of the system and molecular number density. DFT can successfully model vapor–liquid and liquid–solid nucleation and explain phenomena such as spinodal decomposition [9,10]. To correct the sharp interface of the classical droplet model, the diffuse interface theory (DIT) considers a strong dependence of the surface tension on the curvatures, enthalpies, and entropies of bulk solids and liquids within the interface domain [11]. Another important theory is the diffuse interface field theory (DIF), which considers several order parameters that distinguish between

initial and final phases. This theory can be used to predict nucleation in hydrocarbon liquids, liquid metals, and oxide glasses. Based on the experimental evidence from nucleation events in crystallizing solutions, crystals are inferred to be formed through the establishment of clusters with a high-density entities and their structural reorganization. The formation of crystals involves a temporal separation phase necessitating at least two order parameters density and structure for differentiation [12–14].

The described theories employ parameters to solve nucleation problems but cannot directly assess the fundamental thermodynamic properties. Despite its importance, CNT remains incomplete; therefore, there is a need to develop a new nucleation theory solely based on thermodynamic variables to fully comprehend equilibrium dislocations during phase nucleation under high thermal gradients without introducing any process-aiding parameters. The formulation of phase nucleation analyzed in this study is based on a recent work by Ferreira et al. [15,16], starting with its fundamentals and principles that enable the evaluation of thermodynamic variables in phase nucleation. In 2022, Ferreira [17] noted that standard descriptions of any isotherm temperature for multicomponent alloy systems can be expressed in terms of solute concentrations and other variables, which have been widely used in numerical solutions of solidification owing to microsegregation patterns. Recently, Ferreira et al. [18] verified the dependence of molar specific heat capacity of solids on high-temperature gradients through differential thermal analysis/differential scanning calorimetry (DTA/DSC) experiments. The enthalpy and specific heat are known to

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Table 1

– Thermophysical properties of Al-Cu-Si-Fe alloy.

Properties	Symbol	Units	Al-6 wt. %Cu-4 wt. %Si-0.11 wt%Fe
Liquidus temperature	T_L	K	890.65
Solidus temperature	T_{EUT}	K	797.85
Si precipitation temperature	T_{Si}	K	822.49
Liquidus slope	m_L^{Cu}	$K(wt\%Cu)$	−3.700
	m_L^{Si}	$K(wt\%Si)$	−7.422
	m_L^{Fe}	$K(wt\%Fe)$	−2.919
Activation energy (Al liquid)	Q_d^{CuL}	$kJ\ mol^{-1}$	24
	Q_d^{SiL}	$kJ\ mol^{-1}$	30
	Q_d^{FeL}	$kJ\ mol^{-1}$	35
Diffusion constant (Al liquid)	D_0^{CuL}	$m^2\ s^{-1}$	3.0×10^{-9}
	D_0^{SiL}	$m^2\ s^{-1}$	1.0×10^{-9}
	D_0^{FeL}	$m^2\ s^{-1}$	2.34×10^{-7}
Activation energy (Al FCC)	Q_d^{Cu}	$kJ\ mol^{-1}$	130.2
	Q_d^{Si}	$kJ\ mol^{-1}$	137.0
	Q_d^{Fe}	$kJ\ mol^{-1}$	183.4
Diffusion constant (Al FCC)	D_0^{Cu}	$m^2\ s^{-1}$	2.9×10^{-5}
	D_0^{Si}	$m^2\ s^{-1}$	2.48×10^{-4}
	D_0^{Fe}	$m^2\ s^{-1}$	5.3×10^{-3}
Latent heat of fusion (FCC _{Al})	ΔH_{FCC-Al}	$J\ kg^{-1}$	260300
Latent heat of fusion (SILICON)	ΔH_{Si}	$J\ kg^{-1}$	50000
Latent heat of fusion (EUTECTIC)	ΔH_{Eut}	$J\ kg^{-1}$	72700
Low thermal gradient surface tension	σ_0	$N\ m^{-1}$	−1.09
Low thermal gradient surface energy	γ_0	$J\ m^{-2}$	0.183
Dendrite tip growth rate upward	v_L	$mm\ s^{-1}$	$5P^{-0.41}$
Dendrite tip growth rate horizontal	v_L	$mm\ s^{-1}$	$1.2P^{-0.27}$
Local solidification time upward	t_{SL}	s	$3.6 \times 10^{-3} P^{2.32}$
Local solidification time horizontal	t_{SL}	s	$3.58 P^{1.09}$

depend on the cooling rate. The authors noted that the most complete thermodynamic theory of nucleation in literature cannot describe phase nucleation solely using its variables; generally, additional parameters are required to predict nucleation in practical applications. Furthermore, equilibrium dislocations are often neglected in phase nucleation models. By considering the definition of a thermal field in terms of the total system energy, the Gibbs–Thomson coefficient Γ has been redefined in a generalized form known as the thermal field tensor Γ , which allows the phase nucleation to be accurately predicted in terms of the first- and second-order critical radius r_C [18]. By contrast, Rappaz et al. [19] developed a model to predict the dependence of the nucleation radius on the Gibbs–Thomson coefficient, temperature gradients, concentration gradients, and the Peclet number based on Ivantsov's functions. Ferreira confirmed these dependencies of the Gibbs–Thomson coefficient by analyzing the effects of solidification kinetics on the Gibbs–Thomson coefficient in relation to the solute diffusion length scale using Rappaz and Boettinger's secondary dendrite arm spacing (SDAS) model [20] to derive an approach for unsteady solidification [21].

In this study, a novel theory of phase nucleation was elucidated, the thermal field was defined, and the thermal field tensor was derived and applied to the FCC_{Al} phase to evaluate its solution capabilities in relation to the fundamentals of thermodynamic variables.

2. Thermodynamic formulation of phase nucleation

This paper assumes the convention

$$\frac{\partial(\Delta S_V \Delta T f(\theta))}{\partial r} = \left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) f(\theta) + \Delta S_V \Delta T \frac{\partial f(\theta)}{\partial r} \text{ instead of } \frac{\partial(\Delta S_V \Delta T f(\theta))}{\partial r} = \left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) f(\theta) + \Delta S_V \Delta T \frac{\partial f(\theta)}{\partial r} \text{ for deriving equations and formulations based on a previous study [21] for enhanced readability.}$$

As proposed earlier [22], the Gibbs–Thomson equation represents a limiting case of equilibrium homogeneous nucleation considering a newly created and nondeformed surface area A [23], which can only be applied for low thermal gradients, such as those encountered in the production of monocrystals, and is used to calculate the melting point depression as

$$\Delta T = \frac{2\gamma_0}{r_C \Delta S_V} = \frac{2\Gamma}{r_C}, \quad (1)$$

where γ_0 is the surface energy, ΔS_V is the bulk entropy, ΔT is the melting point depression, and r_C is the critical radius. The Gibbs–Thomson coefficient Γ is given in m.K. In such condition, it implies that the undercooling $\Delta T \rightarrow 0$, the temperature gradient $\nabla T \rightarrow 0$, yielding an infinite critical radius $r_C \rightarrow +\infty$. In this case, it is well-known that a sphere is a solid comprising a maximum volume and minimum surface area. The radius is infinite so that the bulk Gibbs-Free energy is lowest possible while the Surface Gibbs-free energy is neglected, i.e., $\lim_{r_C \rightarrow +\infty} \frac{G_S}{r_C} = \lim_{r_C \rightarrow +\infty} \frac{4\pi r_C^2 \gamma_0}{4\pi r_C^3 \Delta S_V} = 0$.

Under equilibrium condition phase grows indefinitely so that a regular crystal can be formed, and the surface area of the solid in a position of a infinite radius. The conclusion is that for a regular crystal to nucleate, the surface area must be in a position infinitely far for the bulk to be assumed as regular.

The dependence of transformation isotherms on pressure, chemical species, melting temperature, work, and other system variables can be determined by considering a new general physical entity called the thermal field T defined according to the first law of thermodynamics, $Q - W = E = \sum_{i=1}^n E_i = U + K + P + \dots + \Sigma_{other}$, which affects the transformation temperature [17,18], with the aim of subsequently redefining the Gibbs–Thomson scalar coefficient Γ as a novel tensor physical entity Γ ,

$$T = \frac{\partial T}{\partial E} E = \frac{\partial T}{\partial U} U + \frac{\partial T}{\partial K} K + \frac{\partial T}{\partial P} P + \frac{\partial T}{\partial W} W + \dots + \frac{\partial T}{\partial \Sigma_{other}} \Sigma_{other}, \quad (2a)$$

$$T = \frac{\partial T}{\partial E} E = T_F - \sum_{j=1}^n \frac{\partial T}{\partial n_i} n_i + \frac{\partial T}{\partial \sigma} \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \Sigma_{other}, \quad (2b)$$

where U , K , P , and W represent internal energy, kinetic energy, potential energy, and work, respectively. σ denotes the normal stress, T_F signifies a transformation temperature, n_i represents the number of moles of species i , and Σ_{other} represents other forms of energy. Upon deriving the thermal field gradient ∇T , we obtain

$$\begin{aligned} \nabla T &= \nabla \left(\frac{\partial T}{\partial E} E \right) = \nabla \left(\frac{\partial T}{\partial E} \right) E + \frac{\partial T}{\partial E} \nabla E \\ &= - \sum_{j=1}^n \nabla \left(\frac{\partial T}{\partial n_i} \right) n_i + \nabla \left(\frac{\partial T}{\partial \sigma} \right) \sigma + \dots + \nabla \left(\frac{\partial T}{\partial \Sigma_{other}} \right) \Sigma_{other} + \nabla T \\ &\quad - \sum_{j=1}^n \frac{\partial T}{\partial n_i} \nabla n_i + \frac{\partial T}{\partial \sigma} \nabla \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \nabla \Sigma_{other}. \end{aligned} \quad (3)$$

The Gibbs–Thomson coefficient can be correctly interpreted by considering a temperature gradient normal to the new surface created. In a general case, this gradient incorporates all components of system energy E , assuming the definition of the thermal field for nonequilibrium conditions regarding the newly created and deformed surface area [24], as proposed in the literature [17,18]. This includes the definition of the thermal field tensor Γ :

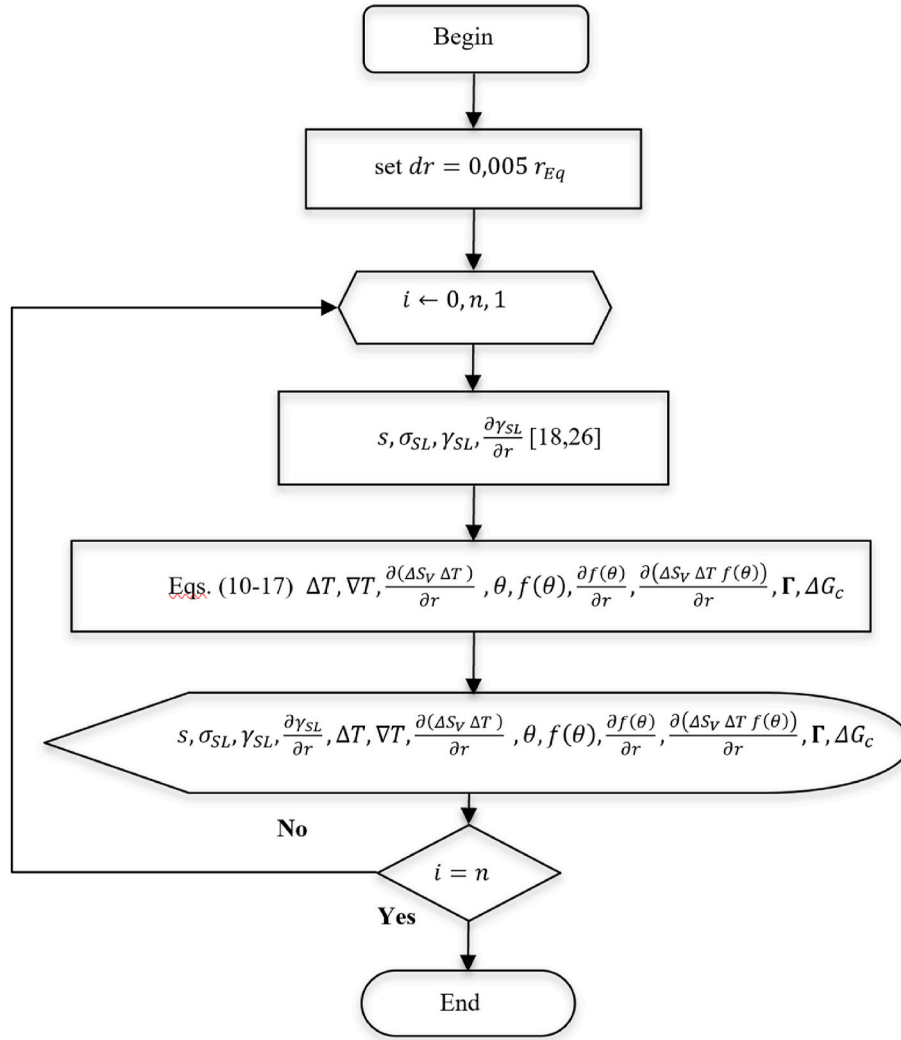


Fig. 1. Diagram illustrating the solution scheme for nucleation variables.

$$\Gamma = \mathbf{A} \bullet \nabla T = \mathbf{A} \bullet \nabla \left[\frac{\partial T}{\partial E} E \right] = \mathbf{A} \bullet \left[\nabla \left(\frac{\partial T}{\partial E} E \right) + \frac{\partial T}{\partial E} \nabla E \right]. \quad (4)$$

The relationships between the thermodynamic variables required to calculate the tensor Γ are assumed using a critical vector radius $\mathbf{r}_c$. The Gibbs critical free energy can be expressed as a function of bulk and surface energies.

$$\Delta G_{\text{Het}} = \left(\frac{1}{3} \pi \mathbf{r}_c^3 \Delta S_V \Delta T + \pi \mathbf{r}_c^2 \gamma_{SL} \right) (2 - 3 \cos \theta + \cos^3 \theta) = \left(\frac{1}{3} \pi \mathbf{r}_c^3 \Delta S_V \Delta T + \pi \mathbf{r}_c^2 \gamma_{SL} \right) f(\theta), \quad (5)$$

where $f(\theta) = 2 - 3 \cos \theta + \cos^3 \theta$.

Consider that the signal of the Gibbs free energy $\Delta G_V = \Delta S_V \Delta T$ is obtained from physics. For example, in the context of homogeneous nucleation, $\mathbf{r}_c$ is expressed as $\mathbf{r}_{c,\text{Hom}} = \frac{2}{\Delta T} \Gamma_{\text{Hom}} = \frac{2}{\Delta T} \mathbf{A}_{\text{Hom}} \bullet \nabla T$. When deriving with respect to the critical radius $\mathbf{r}_c$ to determine the maximum of $\Delta G_{c,\text{Het}}$, the following parabolic equation is obtained:

and in terms of the bulk free energy, $\Delta G_V = \Delta S_V \Delta T$,

$$\frac{\partial(\Delta G_V f(\theta))}{\partial \mathbf{r}} \mathbf{r}_c^2 + 3 \left[\left(\Delta G_V + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right] \mathbf{r}_c + 6 \gamma_{SL} f(\theta) = 0. \quad (7)$$

A first-order nucleation formulation can be derived by assuming only the first-order parabolic description of the critical radius $\mathbf{r}_c$,

$$\mathbf{r}_{c,\text{Het}}^{\text{1st-order}} = \frac{2 \gamma_{SL}}{\Delta T \left[\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}, \quad (8)$$

$$\Gamma_{\text{Het}}^{\text{1st-order}} = \frac{\gamma_{SL}}{\left[\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}, \quad (9)$$

where ΔS_V , $\Delta S_S = \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r}$, and $\Delta S_C = \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}$ represent the bulk, surface, and configurational entropies, respectively. These expressions adhere to the second law of thermodynamics for the thermal field Γ and

$$\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) f(\theta) + \Delta S_V \Delta T \frac{\partial f(\theta)}{\partial r} \right] \mathbf{r}_c^2 + 3 \left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) f(\theta) + \gamma_{SL} \frac{\partial f(\theta)}{\partial r} \right] \mathbf{r}_c + 6 \gamma_{SL} f(\theta) = 0, \quad (6)$$

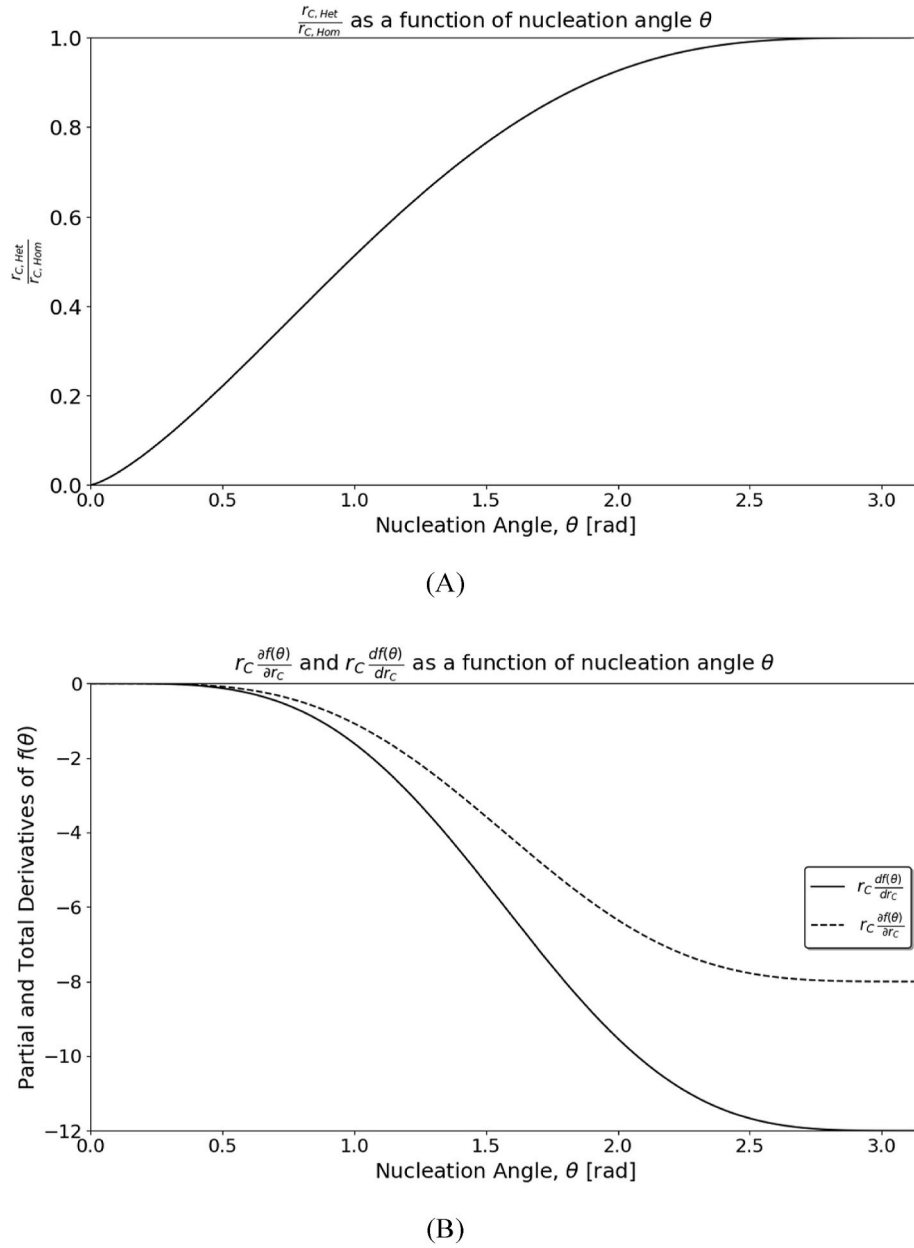


Fig. 2. Dependencies of the critical radius r_C , total derivative $\frac{df(\theta)}{dr_C}r_C$, and partial derivative $\frac{df(\theta)}{dr_C}r_C$ of $f(\theta)$ on the nucleation angle θ : (A) nucleation heterogeneous and homogeneous radii $r_{C,Het}/r_{C,Hom}$ against the nucleation angle θ , (B) derivatives of $f(\theta)$ $\frac{df(\theta)}{dr_C}r_C$ and $\frac{d^2f(\theta)}{dr_C^2}r_C$, (C) first-order angle θ , and (D) second-order angle θ .

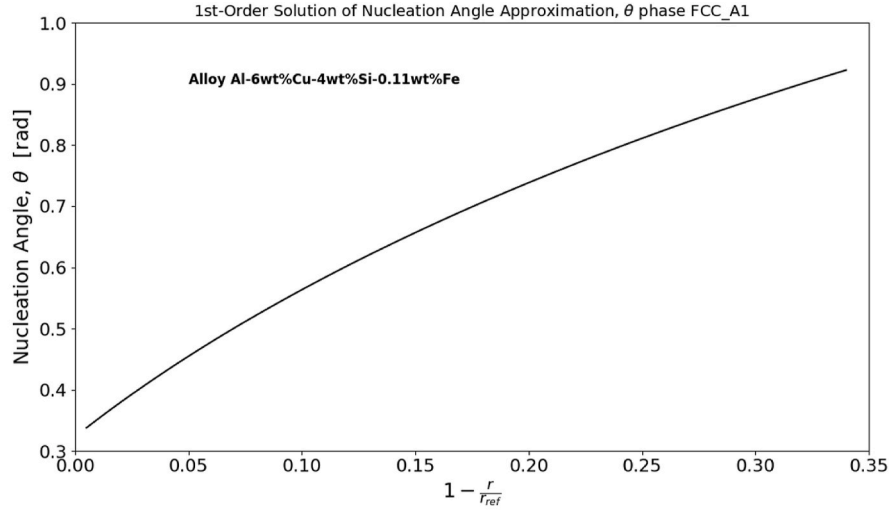
critical nucleation radius r_C tensors.

A second-order radius $r_{C,Het}$ formulation can be derived from maximizing the parabola in Eq. (6):

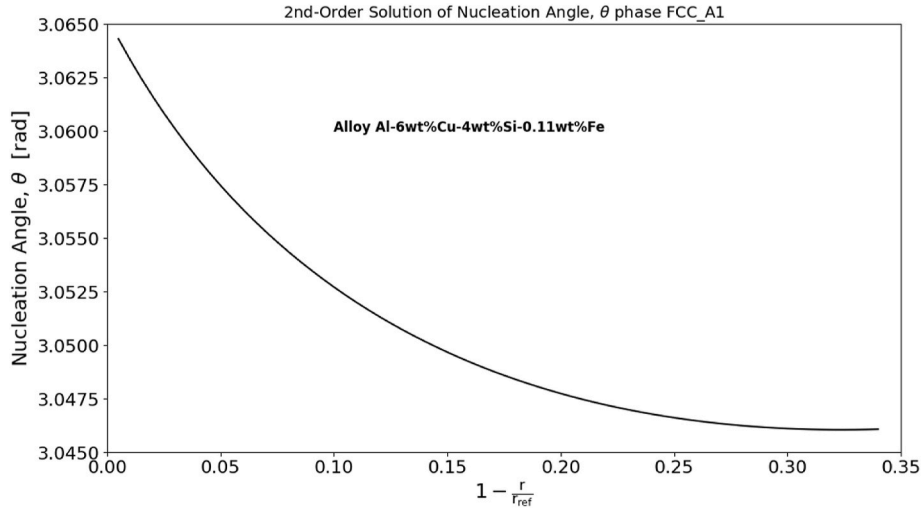
$$\mathbf{r}_{C,Het}^{2nd-order} = -\frac{3}{2} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\left(\frac{\partial \Delta S_V}{\partial r} \Delta T + \Delta S_V \frac{\partial \Delta T}{\partial r} \right) + \Delta S_V \Delta T \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{3}{2} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\frac{1}{f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]}, \quad (10)$$

$$\Gamma_{Het}^{2nd-order} = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\left(\frac{\partial \Delta S_V}{\partial r} + \frac{\Delta S_V}{\Delta T} \frac{\partial \Delta T}{\partial r} \right) + \Delta S_V \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]} = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]}. \quad (11)$$

The thermal field tensor Γ for the first- and second-order nucleation radii can be expressed as follows:



(C)



(D)

Fig. 2. (continued).

$$\Gamma_{Het}^{1st-order} = \frac{\gamma_{SL}}{\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r}\right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}} = \mathbf{A} \cdot \nabla T = \mathbf{A} \cdot \nabla \left[\frac{\partial T}{\partial E} E \right] = \mathbf{A} \cdot \left[\nabla \left(\frac{\partial T}{\partial E} E \right) + \frac{\partial T}{\partial E} \nabla E \right], \quad (12)$$

and

$$\Gamma_{Het}^{2nd-order} = -\frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]} = \mathbf{A} \cdot \nabla T = \mathbf{A} \cdot \left[\nabla \left(\frac{\partial T}{\partial E} E \right) + \frac{\partial T}{\partial E} \nabla E \right]. \quad (13)$$

The first-order nucleation critical free energy can be obtained by

substituting Eq. (8) into Eq. (5) as follows:

$$\Delta G_{C,Het}^{1st-Order} = \frac{16\pi}{3\Delta T^2 \gamma_{SL}} \frac{\left[\left(\Delta S_V + \frac{3}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{3}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]^3} \frac{f(\theta)}{4}. \quad (14)$$

By substituting Eq. (10) into Eq. (5), the following second-order free energy can be obtained:

$$\Delta G_{C,Het}^{2nd-Order} = -\frac{9\pi}{8\Delta T^2} \frac{\left\{ -\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] \right\}^2 \left\{ \Delta S_V \left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] - 2\gamma_{SL} \left(\frac{\partial \Delta S_V}{\partial r} + \frac{\Delta S_V}{\Delta T} \frac{\partial \Delta T}{\partial r} \right) + \frac{\Delta S_V}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right\}}{\left[\left(\frac{\partial \Delta S_V}{\partial r} + \frac{\Delta S_V}{\Delta T} \frac{\partial \Delta T}{\partial r} \right) + \frac{\Delta S_V}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]^3} f(\theta), \quad (15)$$

$$\Delta G_{C,Het}^{2nd-Order} = -\frac{9\pi}{8\Delta T^2} \frac{\left\{ -\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] \right\}^2 \left\{ \Delta S_V \left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right] - 2\gamma_{SL} \left(\frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right) \right\}}{\left\{ \frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right\}^3} f(\theta). \quad (16)$$

The nucleation angle θ is defined by the extent to which substrate energy contributes to minimizing surface free energy, influenced by incomplete chemical bonds in the newly created and deformed surface area A . Considering pre-existing phase domain A , nucleating phase domain B , and any substrate C that energetically interacts with the surface of A to form B , the relationship $\gamma_{BC} + \gamma_{BA} \cos \theta = \gamma_{AC}$ holds. This relationship can be expressed as $\theta = \arccos\left(\frac{\gamma_{AC} - \gamma_{BC}}{\gamma_{BA}}\right)$, and it can be calculated [18] from the homogeneous $r_{C,Hom}$ and heterogeneous $r_{C,Het}$ radii as follows:

$$\frac{r_{C,Het}}{r_{C,Hom}} = e^{-\int_{\pi}^{\theta} \frac{\sin(\theta)[1+\cos(\theta)]}{2-\cos(\theta)-\cos^2(\theta)} d\theta}. \quad (17)$$

When the thermal field gradient ∇T is constant, θ for any $\theta < \pi$ is also constant. This indicates that the radii $r_{C,Hom}$ and $r_{C,Het}$ differ from one another but are constant for a given θ . Consequently, a regular spherical cap surface area is created and deformed. However, in a general case, it depends on $\theta = f(\mathbf{r}, \theta, \varphi)$, since $\nabla T = f(\mathbf{r}, \theta, \varphi)$, where $\mathbf{r}$ is a vector radius [17,18].

The components of the thermal field tensor Γ , which influence the temperature, can be inferred based on the work w , molar composition n_i of component i , stress tensor τ , and other forms of energy Σ_{other} as follows:

$$\begin{aligned} \mathbf{A} \cdot \nabla T &= \mathbf{A} \cdot \left[\sum_{j=1}^n \nabla \left(\frac{\partial T}{\partial n_i} \right) n_i + \nabla \left(\frac{\partial T}{\partial \sigma} \right) \right. \\ &\quad \bullet \sigma + \dots + \nabla \left(\frac{\partial T}{\partial \Sigma_{other}} \right) \Sigma_{other} + \sum_{j=1}^n \frac{\partial T}{\partial n_i} \nabla n_i + \frac{\partial T}{\partial \sigma} \\ &\quad \left. \bullet \nabla \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \nabla \Sigma_{other} + \nabla T \right], \end{aligned} \quad (18)$$

which provides the first-order approximation $r_C^2 \approx 0$ for the nucleation formulation expression,

$$\begin{aligned} \frac{\gamma_{SL}}{\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}} &= \hat{n} \cdot \left[\sum_{j=1}^n \nabla \left(\frac{\partial T}{\partial n_i} \right) n_i + \nabla \left(\frac{\partial T}{\partial \sigma} \right) \right. \\ &\quad \bullet \sigma + \dots + \nabla \left(\frac{\partial T}{\partial \Sigma_{other}} \right) \Sigma_{other} + \sum_{j=1}^n \frac{\partial T}{\partial n_i} \nabla n_i + \frac{\partial T}{\partial \sigma} \\ &\quad \left. \bullet \nabla \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \nabla \Sigma_{other} + \nabla T \right] A(r_C), \end{aligned} \quad (19)$$

and the second-order nucleation radius,

$$\begin{aligned} \frac{3}{4} \frac{\left[\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}{\left[\frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r} \right]} &= \hat{n} \cdot \left[\sum_{j=1}^n \nabla \left(\frac{\partial T}{\partial n_i} \right) n_i + \nabla \left(\frac{\partial T}{\partial \sigma} \right) \right. \\ &\quad \bullet \sigma + \dots + \nabla \left(\frac{\partial T}{\partial \Sigma_{other}} \right) \Sigma_{other} + \sum_{j=1}^n \frac{\partial T}{\partial n_i} \nabla n_i + \frac{\partial T}{\partial \sigma} \\ &\quad \left. \bullet \nabla \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \nabla \Sigma_{other} + \nabla T \right] A(r_C). \end{aligned} \quad (20)$$

The nucleation rate I is typically expressed as dependent on temperature [25]. This formulation includes the thermal gradient and undercooling, expressed in terms of equilibrium:

$$I = \frac{D A}{\lambda^4} \frac{N}{V} \exp\left(\frac{\Delta G_C}{\Delta G_{C,Eq}}\right), \quad (21)$$

where D is the self-diffusion coefficient in m^2s^{-1} , λ is the distance an atom must jump from a phase to the nucleating new surface area in m , A is the surface area of the new phase in m^2 , $\frac{N}{V}$ is the density of modes in the new phase in m^{-3} [18], $\Delta G_{C,Eq}$ is the equilibrium critical free energy in J for a new created surface area, and ΔG_C is the equilibrium critical free energy in J for any newly created and deformed surface area.

The solution for the surface stress tensor [24], utilized for computing the surface energy [26] of an isotropic medium, is detailed in Ref. [27].

Concerning the absolute equilibrium to provide a reasonable explanation by comparing the first-order radius nucleation approximation with the classical equation for heterogeneous nucleation. The classical equilibrium radius is given by,

$$r_C^{classical} = \frac{2\gamma_{SL}}{\Delta T \Delta S_V}$$

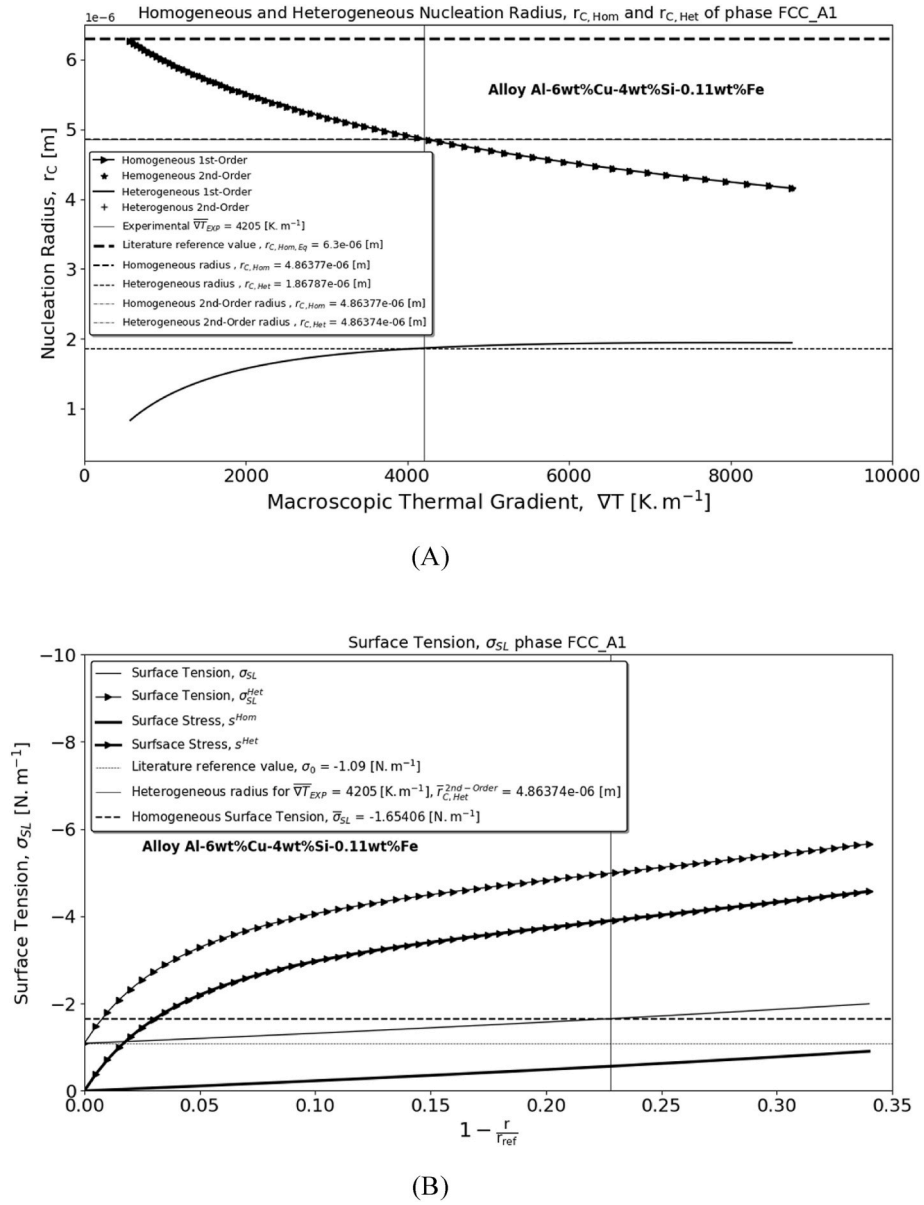


Fig. 3. Approximation of the first-order and exact second-order nucleation variables in terms of the decrease in dimensionless radius and macroscopic thermal gradient: (A) critical radius r_C , (B) surface tension σ_{SL} and surface stress s , (C) surface energy γ_{SL} , and (D) partial derivatives of surface energy $\frac{\partial \gamma_{SL}}{\partial r}$.

as the classical homogeneous and heterogeneous radii are given by the same equation for r_C , and both are valid only for $\Delta T \rightarrow 0$, $\nabla T \rightarrow 0$ and $r_C \rightarrow +\infty$ in order to $\lim_{r_C \rightarrow +\infty} \frac{G_S}{\Delta G_V} \frac{4\pi r_C^2}{3\pi r_C^3} = \lim_{r_C \rightarrow +\infty} \frac{3\gamma_{SL}}{\Delta G_V} \frac{1}{r_C} \rightarrow 0$.

The current study, for any finite nucleation radius, from a first-order nucleation approximation perspective,

$$\mathbf{r}_{C,Het}^{1st-order} = \frac{2\gamma_{SL}}{\Delta T \left[\left(\Delta S_V + \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \right) + \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \right]}$$

where, two additional terms appeared, the surface entropy $\Delta S_S = \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r}$ and the substrate contribution entropy $\Delta S_C = \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}$. As the last term has to do with the change in orientation, it is called configurational entropy. However, in the limit, when $r_C \rightarrow +\infty$, $\Delta S_S = \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r} \rightarrow 0$, $\Delta S_C = \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r} \rightarrow 0$, $\theta \rightarrow \pi$ and $\mathbf{r}_{C,Het}^{1st-order} \rightarrow \mathbf{r}_C^{classical}$, causing homogeneous

nucleation to occur.

2.1. Important Considerations on entropy

After deriving the framework of phase nucleation, some important aspects of bulk entropy should be addressed. Boltzmann derived entropy considering the total number in a cluster as Ω ,

$$S = k \ln \Omega, \quad (22)$$

where k is the Boltzmann constant.

The sum of all statistics weights provides,

$$\Omega = \sum_i w_i = \sum_i \frac{N!}{\prod_i N_i!}, \quad (23)$$

and N is the number of different particles distributed along N_i energy

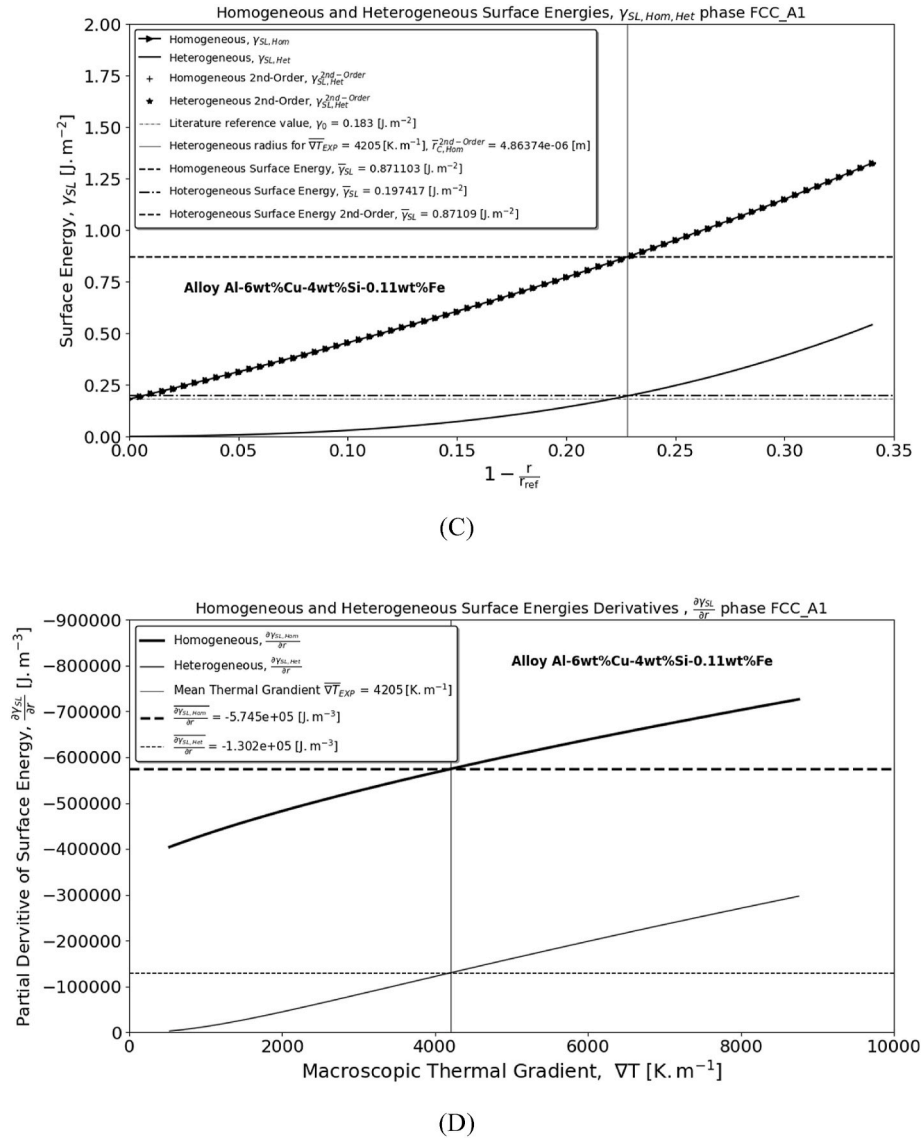


Fig. 3. (continued).

states.

The total weight Ω can be expressed in terms of gamma function $\Gamma(z)$.

$$\Omega = \sum_i \frac{N!}{\prod_i N_i!} = \sum_i \frac{\Gamma(N+1)}{\prod_i \Gamma(N_i+1)} = \sum_i \frac{N\Gamma(N)}{N_i\Gamma(N_i)!} = \sum_i \frac{N\Gamma(N)}{\Gamma(\Gamma(N_i+1)+1)}, \quad (24)$$

and by substituting Eq. (24) into Eq. (22),

$$S = k \ln \sum_i \frac{N!}{\prod_i N_i!} = k \ln \sum_i \frac{N\Gamma(N)}{N_i\Gamma(N_i)!} = k \ln \sum_i \frac{N\Gamma(N)}{\Gamma(\Gamma(N_i+1)+1)}, \quad (25)$$

The entropy for the most likely state ω_{max} can be derived as

$$S \approx k \ln \omega_{max} \approx k \ln \frac{N!}{\prod_i N_i!} \approx k \ln \frac{N\Gamma(N)}{\Gamma(N_i+1)!} \approx k \ln \frac{N\Gamma(N)}{\Gamma(\Gamma(N_i+1)+1)}, \quad (26)$$

According to Eq. (26), any further entropy derivation must refrain from employing the Stirling approximation and delve into a more comprehensive mathematical description based on the gamma function

relationships. The author holds the belief that this description must encompass the domain of physical solutions for thermodynamics-dependent problems, which are currently only achieved through mathematical physics methods. The disadvantage is traceable to masking, disguising, and obscuring the clarity of the physical solutions deriving from the theoretical relations for thermodynamic properties and transport equations about those obtained from pure mathematical solutions of PDEs in the hope of improving theoretical predictions and elucidating the Third Law of Thermodynamics.

2.2. Application to the universe scale

The thermal field tensor Γ is derived from the first and the second laws of Thermodynamics, as shown in equations (2)–(4). It comprises two fundamental laws of nature, which can be applied to the cosmic scale by expressing the RHS of Equation (4) through Einstein's stress-energy tensor in the first law term. The LHS, which represents the second law, must be written by the surface stress-energy tensor determined by relativistic mechanics. Suppose Eq. (4) is a law of nature. In this case, the observed effects of dark energy and dark matter [28–30] must only

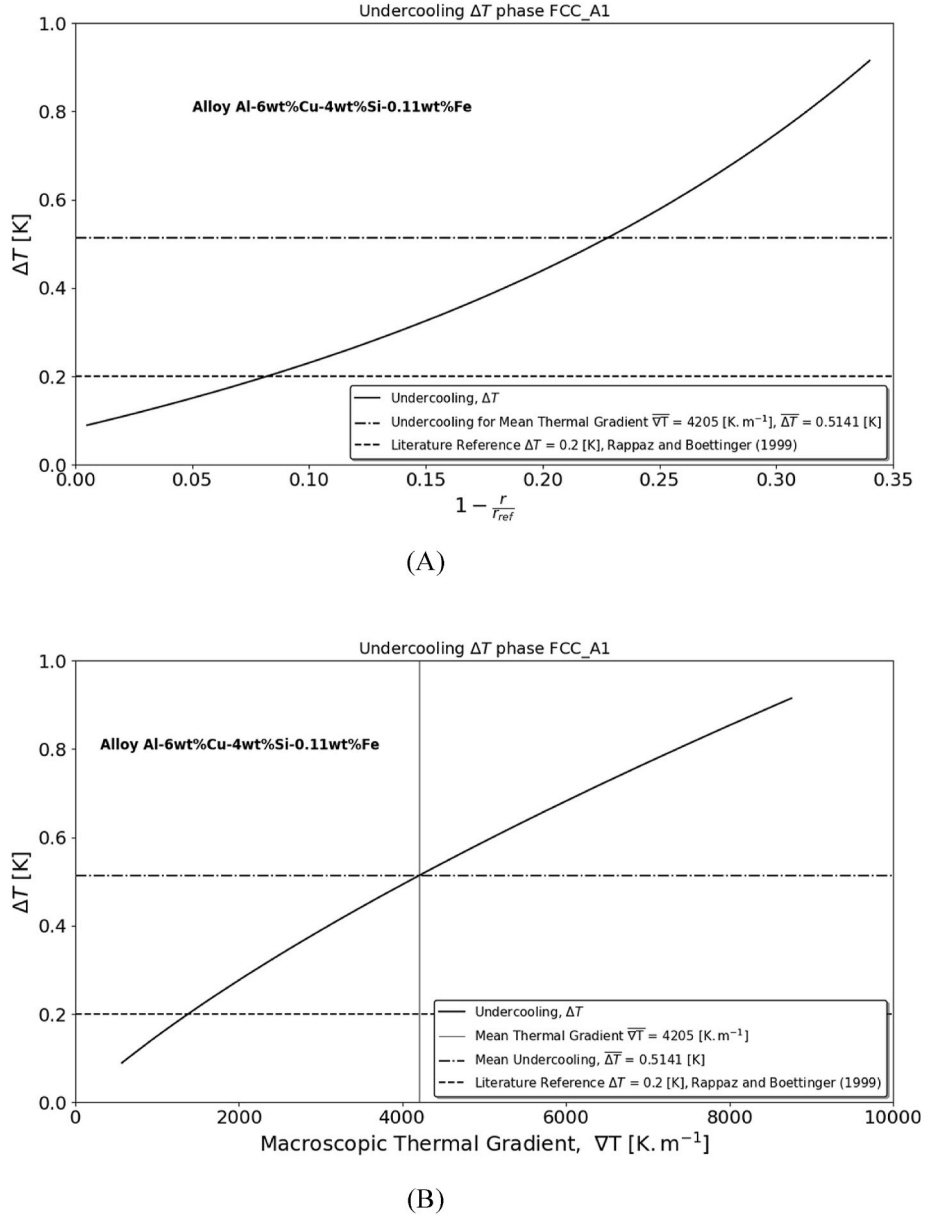


Fig. 4. Nucleation variables: (A) undercooling ΔT as a function of the decrease in dimensionless radii, (B) undercooling ΔT as a function of the thermal gradient, (C) derivative of bulk Gibbs free energy, and (D) heterogeneous first- and second-order Gibbs–Thomson Γ against microscopic thermal gradient.

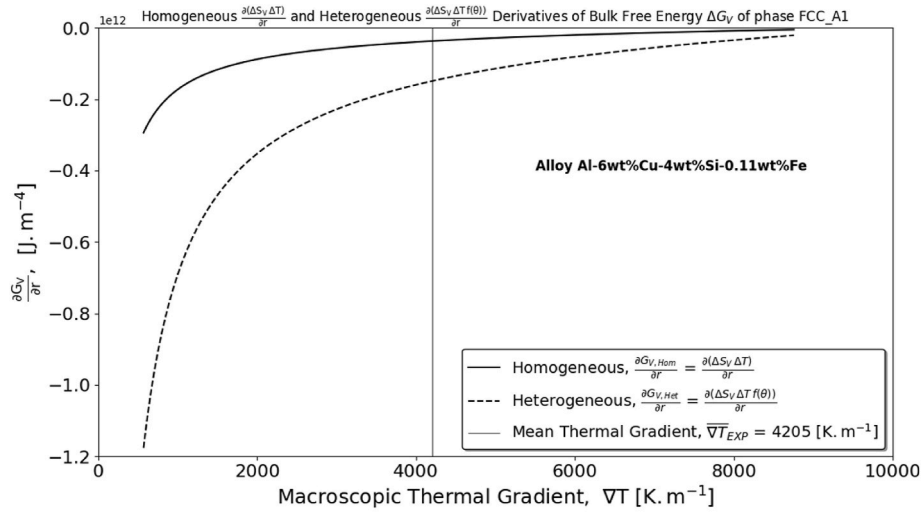
be a consequence of spacetime expansion due to mass condensation related to the decrease in the universe's temperature. These effects can be predicted by relativistic mechanics, as found in Ref. [25].

Let's construct the full thermodynamic image of the universe. At its initial stage, when the temperature begins to decrease, condensation of primordial particles, e.g. leptons in leptogenesis as other particles, begins to condense in a process called light particle condensation. As the temperature continued to decrease, light atoms formed, allowing the agglomeration of matter along certain paths and establishing the nuclear fusion of the first stars, i.e. heavy atoms condensation. In these agglomerates, matter condenses into more complex atoms triggered by nuclear fusion, which causes spacetime to deform continuously as temperature drops. The decrease in temperature can be understood as an increase in undercooling related to any transformation, including fusion, which speeds up the Universe's expansion. In the same way as a monocrystal, the bulk entropy means equilibrium, and the surface

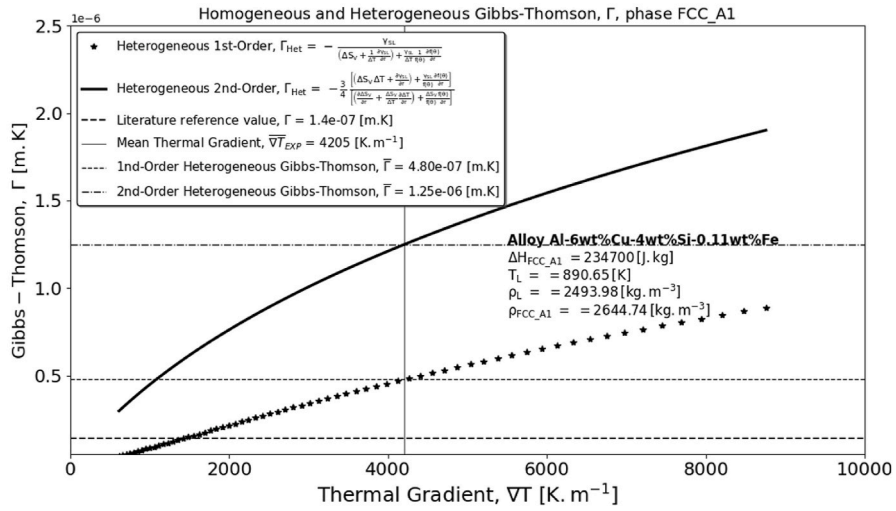
means non-equilibrium as previously derived. The radius will trend to the infinite, so the bulk trends to the equilibrium. What does it mean? At the final stage of the universe, the most probable energy state in the bulk will be the lowest, and the Universe will attain equilibrium, albeit in a state of complete coldness and darkness.

3. Results and discussion

The results presented in this section correspond to the solution of the FCC_A1 phase nucleation considering the properties found in Table 1, where nucleation variables are determined as a function of decreasing radius. The solution begins by considering the equilibrium radius r_{Eq} , establishing $dr = 0.005 r_{Eq}$, and making $r = r_{Eq} - i dr$. The surface stress s , surface tension σ_{SL} , surface energy γ_{SL} , and its derivative $\frac{\partial \gamma_{SL}}{\partial r}$ are computed, as these quantities depend solely on the nucleation radius. This paper employs a first-order approximation solution to compute ΔT



(C)



(D)

Fig. 4. (continued).

and ∇T and a second-order formulation to calculate the second-order nucleation-dependent variables, such as $\frac{\partial(\Delta S_V \Delta T)}{\partial r}$, $f(\theta)$, $\frac{\partial f(\theta)}{\partial r}$, and $\frac{\partial(\Delta S_V \Delta T f(\theta))}{\partial r}$, as presented in Fig. 1.

In Fig. 2A, the ratio of heterogeneous to homogenous radii is expressed in terms of the nucleation angle θ . In the case of homogeneous nucleation where $\theta = \pi$ and for $\theta < \pi$, the substrate contribution to the critical free energy ΔG_C increases, resulting in a decrease in free energy compared to homogeneous free energy. Note that another mechanism contributes to the decrease in surface energy in multicomponent systems, where the solute concentration at the newly created surface changes to decrease the surface free energy. In Fig. 2B, the partial and total derivatives of $f(\theta)$ with respect to the radius r_C are plotted against the nucleation angle θ , highlighting the knowing of $\frac{\partial f(\theta)}{\partial r_C}$ for heterogeneous nucleation. Fig. 2C and D shows the first- and second-order nucleation angles θ separately plotted as a function of a decrease in the dimensionless radius. A first-order solution serves as an approximation for nucleation, where $r_C^2 \approx 0$.

In the study of FCC_A1 phase nucleation for the Al-6Cu-4Si-0.11Fe

alloy (in wt.%), nucleation variables are calculated considering a macroscopic thermal gradient ∇T typical of a DTA/DSC device or any transient DC solidification setup. In Fig. 3A, homogeneous and heterogeneous radii are plotted for both first-order nucleation approximation and second-order formulations against the thermal gradient ∇T . All radii fit one another, except $r_{C, Het}^{1st}$. The first-order nucleation radius (Eq. (8)) is an approximate formulation of the linear term of the parabola (Eqs. (6) and (7)), whereas the second-order formulation provides the exact radius. Fig. 3C illustrates the surface stress s and surface tension σ_{SL} plotted against a decrease in the dimensionless radius for homogeneous and heterogeneous nucleation. After calculating the surface stress s for homogeneous nucleation, the heterogeneous surface stress is not used in any variable determination, as heterogeneous surface energy is derived from the nucleation formulation $\gamma_{SL, Het} = \gamma_{SL, Hom} f(\theta)$ and not from heterogeneous surface stress. Fig. 3C and D shows the surface energy $\gamma_{SL, Het}$ and its derivative $\frac{\partial \gamma_{SL}}{\partial r}$ in terms of the macroscopic thermal gradient ∇T . Homogeneous surface energies are the same for the first- and second-order formulations. However, for heterogeneous nucleation, these parameters differ significantly. Surface energy and the value of its

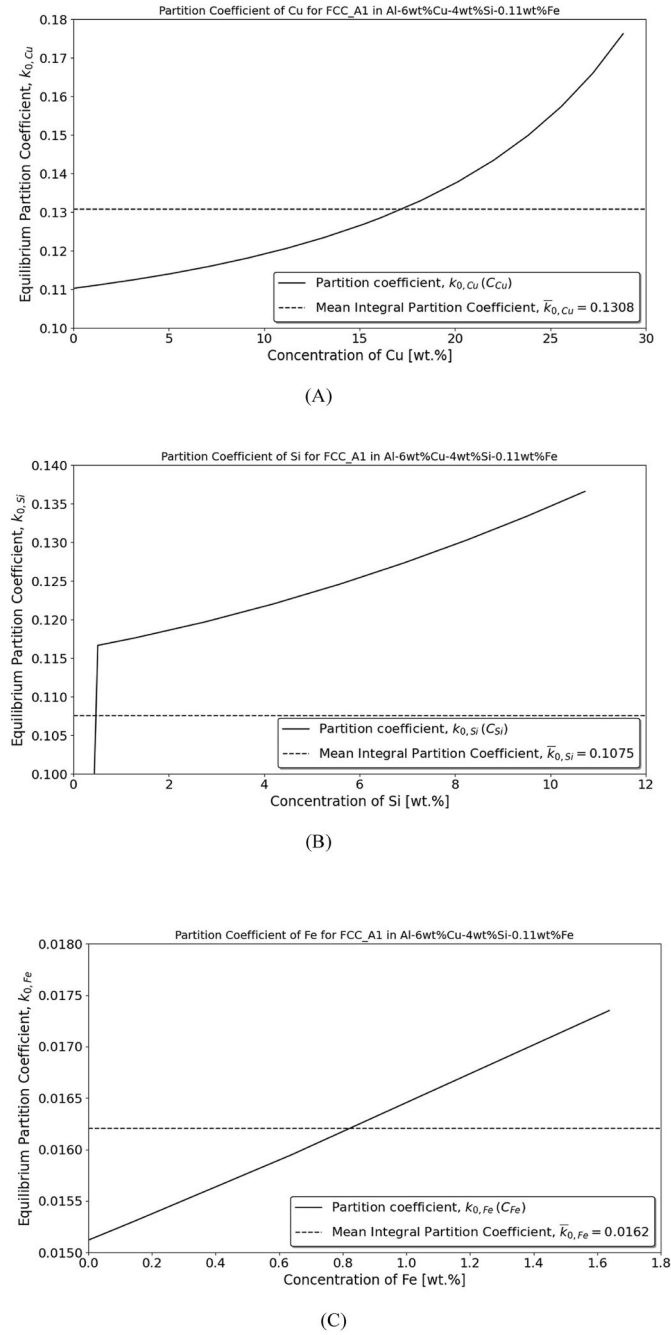


Fig. 5. Equilibrium partition coefficient as a function of the composition between FCC_A1 and LIQUID#1 phases: (A) Cu, (B) Si, and (C) Fe.

derivative increase with increasing thermal gradient.

Fig. 4A and B shows undercooling as a function of decreasing radius and thermal gradient. Note that the undercooling exhibits a down-concave shape, expressed in terms of the decrease and increase in dimensionless radius against the thermal gradient. Fig. 4C shows the derivatives of the volumetric free energy with respect to homogeneous and heterogeneous radii, and the dependence of these curves on the thermal properties was analyzed. Lower values were obtained for heterogeneous nucleation, but these values increased with higher thermal gradients, approaching the values seen in homogeneous nucleation for high thermal gradients. Fig. 4C shows the heterogeneous Gibbs–Thomson coefficients (thermal field tensors) for heterogeneous nucleation, considering first-order approximation and second-order nucleation. The first-order predictions $\Gamma_{Het}^{1st-order}$ are lower than the

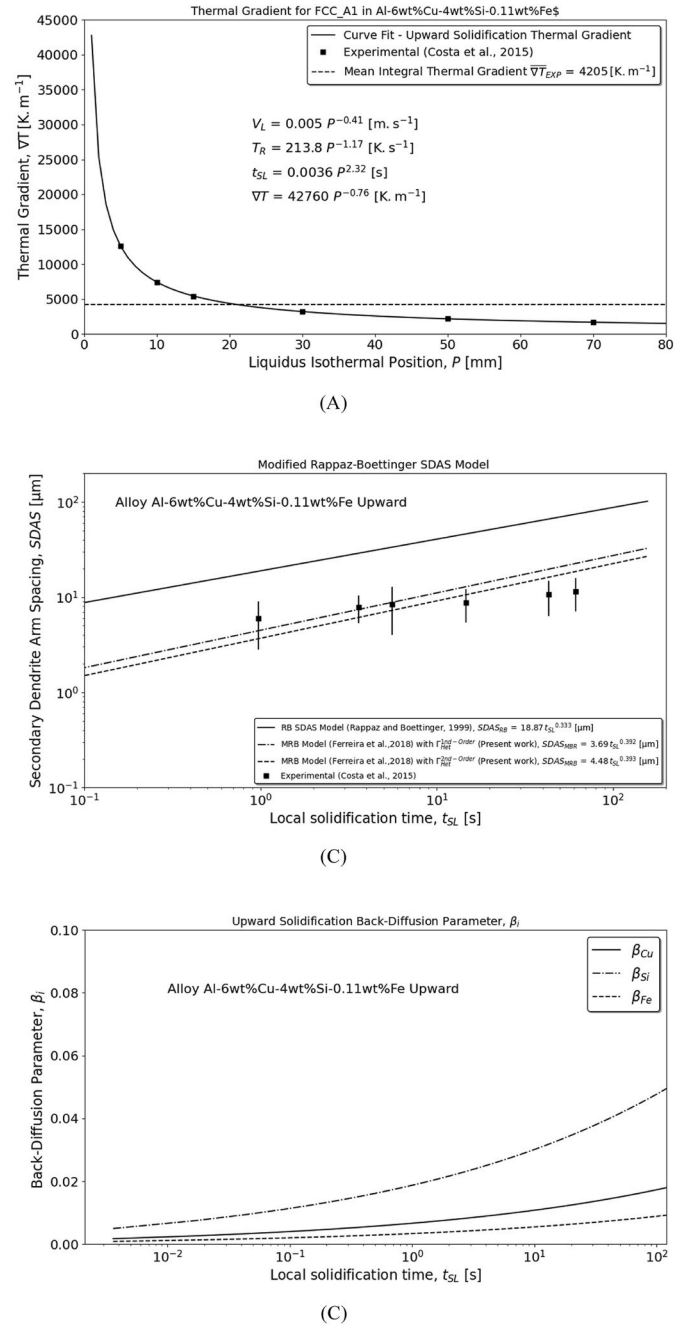


Fig. 6. Upward transient solidification: (A) experimental thermal gradient and mean integral thermal gradient, (B) SDAS prediction model [21], and (C) prediction of back-diffusion parameter β_i .

second-order predictions $\Gamma_{Het}^{2nd-order}$.

The equilibrium dislocation is determined using the thermal field tensor $\Gamma = \mathbf{A} \bullet \nabla T$. An illustrative application involves predicting SDAS [21] for two solidification setups: horizontal and upward. The mean integral thermal gradients are calculated as $\nabla T_{EXP} = 4025$ K m⁻¹ for upward solidification and $\nabla T_{EXP} = 4267,76$ K m⁻¹ for horizontal solidification, using information from a central line thermocouple, which is approximate for horizontal solidification [31]. Equilibrium partition coefficients, liquidus slopes, and phase-change entropies necessary for the SDAS model under nonequilibrium kinetics [21,32] were derived from Thermo-Calc calculations using the TTAL7 Thermotech Aluminum Database v.7. Fig. 5 illustrates the equilibrium partition coefficients for Cu (Fig. 5A), Si (Fig. 5B), and Fe (Fig. 5C) in terms of solute concentrations.

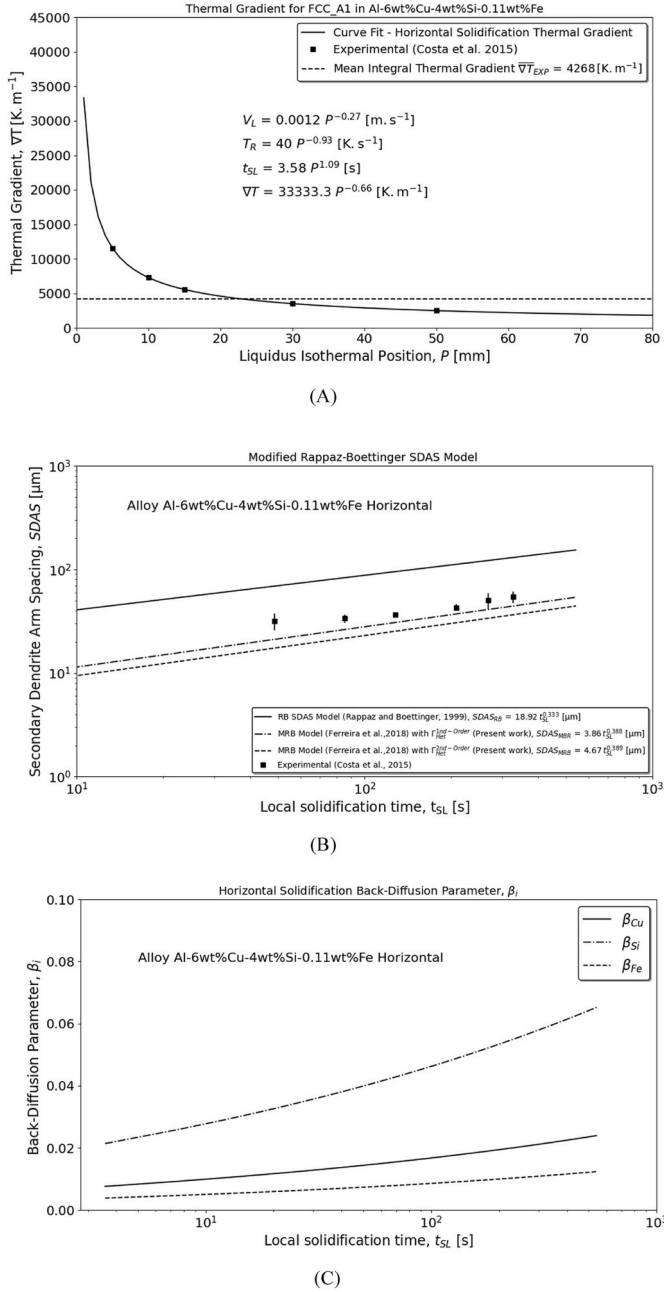


Fig. 7. Horizontal transient solidification: (A) experimental thermal gradient and mean integral thermal gradient, (B) SDAS prediction model [21], and (C) prediction of back-diffusion parameter β_i .

$$SDAS = 5.5 (M \bullet t_{SL})^{\frac{1}{3}}, \quad (27)$$

and

Fig. 5A and 6A show the experimental thermal gradients [33] for upward and horizontal solidifications plotted against the liquidus position. The mean integral thermal gradient was applied to calculate the mean Gibbs–Thomson equation, representing the equilibrium dislocation. Fig. 6B and C and 7B–C show the experimental SDAS and theoretical predictions from the Rappaz and Boettinger SDAS model [20] as well as the modified Rappaz and Boettinger model [21], against local solidification time t_{SL} , considering heterogeneous Gibbs–Thomson coefficients $\Gamma_{Het}^{1st-order}$ and $\Gamma_{Het}^{2nd-order}$. The theoretical predictions agreed with the experimental scatter for both upward and horizontal setups. Notably, for horizontal data, the thermal gradient approximation is influenced by convection because of the non-unidirectional solidification. The back-diffusion parameter β_i is calculated as a function of t_{SL} . Values for all solutes closely correspond to the Scheil equation, indicating finite diffusion. Horizontal solidification results in a higher β_{Si} compared to upward solidification, with $t_{SL} = 550$ s versus $t_{SL} = 100$ s, respectively.

Fig. 8 shows the nucleation rate I prediction, Eq. (21), for the FCC_A1 phase considering $A_{Hom} = 4\pi r_C^2$ and $A_{Het} = 2\pi r_C^2(1 - \cos \theta)$, self-diffusion coefficient D [34,35], density of modes in solid FCC_A1, $\frac{N}{V} = 1, 17594 \times 10^{28} \frac{modes}{m^3}$ [18], and the atom jump distance from the liquid to FCC_A1 phase, $\lambda = 0.25 \times 10^{-9}$ m. High nucleation rates are expected because of the macroscopic measurement of thermal gradients, similar to those observed in DC solidification or DTA/DSC devices. Analysis of the nucleation rate confirms the nature of the thermal gradient measurement and validates the physics of nucleation on a macroscopic scale.

Tables 2, 3 and 4 in the supplementary material refer to the nucleation variables simulated data in terms of the decrease in the homogeneous radius, which allows FCC_A1 based on the Al-6wt%Cu-4wt%Si-0.11 wt%Fe alloy to be evaluated for several thermal gradients. The data in bold are the nucleation variables used in the theoretical prediction of SDAS against the experimental scatter.

4. Conclusions

The following conclusions can be drawn from the results and discussion presented in this paper.

- The exact formulation of phase nucleation follows a parabolic model.
- Any formulation assuming a first-order dependence on nucleation radius can only be considered an approximation of a general formulation.
- The thermal field is defined in accordance with the first law of thermodynamics.
- The thermal field tensor Γ is defined as the thermal gradient normal to the newly created and deformed surface area.
- Fundamental entropies, i.e., the bulk entropy ΔS_V , surface entropy $\Delta S_S = \frac{1}{\Delta T} \frac{\partial \gamma_{SL}}{\partial r}$, and configurational entropy $\Delta S_C = \frac{\gamma_{SL}}{\Delta T} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}$, can be derived under the first-order approximation $r_C^2 \approx 0$.

$$M = \frac{-\Gamma \sum_{j=1}^n w_j \beta_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) (c_{f,j} - c_{0,j}) / D_j} \ln \left[\frac{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{f,j} / D_j}{\sum_{j=1}^n m_j \left(1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right) c_{0,j} / D_j} \right]. \quad (28)$$

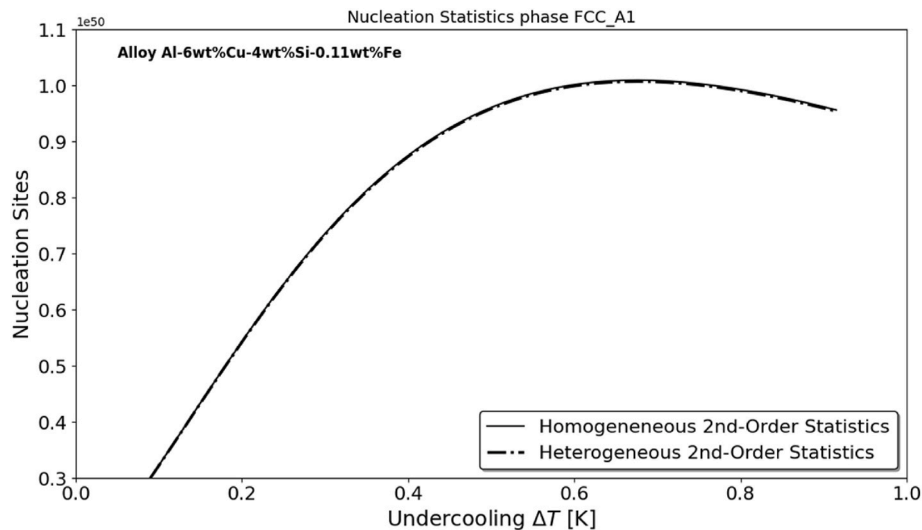


Fig. 8. Nucleation rate for second-order homogeneous and heterogeneous as a function of undercooling.

- The proposed nucleation formulation provides a general framework for nonequilibrium phase transitions and directly relates to statistical thermodynamic properties.
- The second-order formulation is expressed in terms of the derivatives of surface-free energy, bulk-free energy, and configurational entropy.
- Comparison between first- and second-order nucleation angles θ , initially appears contradictory, but exact solutions are only attainable through second-order formulations.
- Lower thermal gradients cause lower undercooling, making the critical radius to infinity and nucleation angles converge toward π , denoting homogeneous nucleation.
- In multicomponent nucleating phases, surface energy is minimized by changing the surface concentration relative to bulk, as predicted by Butler's formulation, and this might be responsible for large second-order nucleation angles close to π .
- The prediction of equilibrium dislocations as a function of the macroscopic thermal gradient based on Gibbs–Thomson coefficient calculations agrees with experimental data for Al–6Cu–4Si–0.11Fe (in wt.%) in upward and horizontal transient solidifications.
- The analysis of high nucleation rates confirms the importance of the thermal gradient measurement for the phase nucleation analysis. the proposed formulation validates the physics of nucleation on a macroscopic scale, which allows it to be applied to phase transformations based on the principle of continuum mechanics and thermophysical properties from statistical thermodynamics and quantum mechanics, such as application in DTA/DSC devices, cryogenics, phase transformation, and weather forecasting models.
- Bulk entropy represents crystalline order and surface entropy disorder for a regular crystal with infinite nucleation radius.
- Any finite crystal cannot be considered regular due to surface entropy affecting crystal regularity.
- About the entropy behaviour on the scale of the universe can be concluded: The universe expands because of matter condensation to minimize the bulk entropy (from fundamental particles at its initial stages towards the more complex matter, from light to heavy atoms). The surface-free entropy affects the regularity trend of the universe's bulk entropy only near its boundaries at infinite positions, which allows us to affirm that the universe will ultimately tend towards the lowest possible energy state in the bulk when $T \sim 0K$, and its surface continue to expand as long as fundamental energy state is not reached. The universe's expansion is accelerated by the continuing decrease in the average temperature towards absolute zero, which

triggers higher rates of matter condensation, deforming the space-time fabric and expanding it.

- It can be inferred that the universe's expansion is caused by the condensation of matter, which reduces the bulk entropy since primordial particles and light atoms in the universe's initial stages to more complex matter triggered by nuclear fusion. The surface-free entropy affects the regularity trend of the universe's bulk entropy only near its boundaries at infinite positions, which allows us to affirm that the universe will ultimately tend towards the lowest possible energy state in the bulk when $T \sim 0K$, and its surface continue to expand as long as fundamental energy state in the bulk is not reached.
- The universe's expansion is accelerated by the continuing decrease in the average temperature towards absolute zero, which results in higher rates of matter condensation, deforming the spacetime fabric and expanding it. The effect attributed to dark matter and dark energy may then be determined by the relativistic mechanics and thermal field tensor, related to the terms of the first law by the stress-energy tensor and the second law by the surface stress-energy tensor.

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CRediT authorship contribution statement

I.L. Ferreira: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.physb.2024.416494>.

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